Supplementary material

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I. Dependence of Power Flexibility Region Boundary Identification on the Initial Point

MOST existing studies that identify the power flexibility region (PFR) boundary via discrete angular scanning start from the origin or an interior point and scan allover to find points that maximize the PFR. However, the correctness of the above algorithm often depends on the initial point. An example is provided to illustrate this.

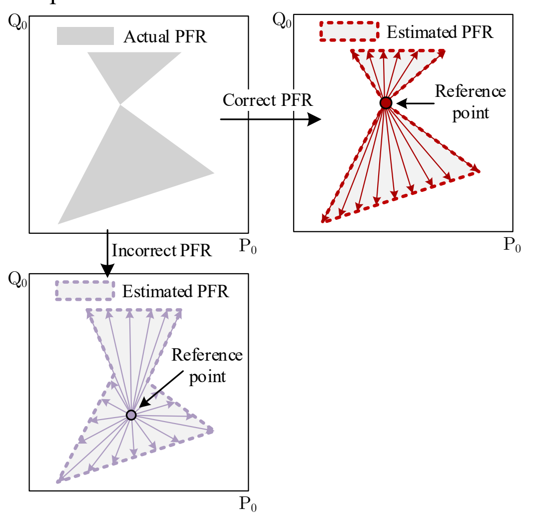

Fig. R1. Examples of the PFR estimation.

In the example shown in Fig. R1, the true PFR is illustrated in Fig. 1(a). By choosing different reference points, typically the origin or a power-flow feasible point, and performing a full-angle scan, the two PFRs shown in Fig. 1(b) and (c) are obtained. It is evident that the identified PFR depends on the choice of the reference point and may be inaccurate.

II. The Pseudocode of the Proposed PFR Identification Algorithm

The complete procedure of the proposed PFR identification algorithm is given in the Algorithm 1.

Algorithm 1: The proposed PFR Identification Algorithm.

Input: The number of scanning angles M ; an empty feasible/infeasible set $\Theta_{\rm fe}/\Theta_{\rm inf}$ and a temporary feasible/infeasible set $\Theta_{\rm fe}^{\rm m}/\Theta_{\rm inf}^{\rm m}$; the gap for terminating the iteration l_{min}.

Output: The feasible set $\Theta_{\rm fe}$ and the infeasible set $\Theta_{\rm inf}$.

```
1  for m = 0: M do        // This step can be executed in parallel.
2    Feasibility Initialization;
3    if the problem is infeasible, then
4      break;
5    else
6      if l_1 − l_2 ≤ l_min, then
7        Θ_fe^m ← Θ_fe^m ∪ {X_1, X_2} and break;
8      else
9        Feasibility Expansion;
10       if l_4 ≥ l_3, then
11         break;
12       else
13         Θ_inf^m ← Θ_inf^m ∪ {X_3, X_4}; Feasibility Exploration;
14         if l_5 = l_4 + l_tol, then
15           Θ_inf^m ← Θ_inf^m \ {X_4}; go to 25;
16         else
17           Θ_inf ← Θ_inf ∪ {X_5, X_6}; go to 25;
18         end
19       end
20     end
21   end
22   Θ_fe ← Θ_fe ∪ Θ_fe^m; Θ_inf ← Θ_inf ∪ Θ_inf^m;
23   m ← m + 1;
24 end
25 if l_3 − l_6 > l_min, then
26   (X_4, Y_4, Z_4) ← (X_6, Y_6, Z_6); l_4 ← l_6; go to 10;
27 elseif l_3 < l_6
28   Θ_inf^m ← Θ_inf^m \ {X_4};
29 else
30   break;
31 end
```

III. Supplementary Descriptions of the Test Model

In this part the supplementary descriptions about relevant parameters of the modified Swiss 13-node, 38-node, and unbalanced 162-node test systems is shown in Table RI-RIII.

TABLE RI
Values of Relevant Parameters for 13-node System (Unit: p.u.)

Name	Values	Name	Values	Name	Values
l_{min}	0.005	v_{MV}^{min}	0.95	S_{PV}^{max}	1
M	120	v_{MV}^{max}	1.05	P_{load}	0
ε	0.001	P_{PV}^{max}	1	Q_{load}	0
S_{Line}	1	P_{PV}^{min}	0.2	$q_{C1/2/3/4}$	0/0.5/1/1.5
v_{OLTC}	0.98/1/1.02	$\tan\phi_{PV}$	1	S_c	1

And l_{min} is the gap for terminating the iteration; M is the number of scanning angles; ε is a judgement parameter in the algorithm; S_{Line} denotes the per-unit value of the maximum line capacity; v_{OLTC} represents the voltage tap positions adjustable by the on-load tap changer; $v_{MV}^{min}/v_{MV}^{max}$ are the upper and lower limits of the nodal voltage, respectively; P_{PV}^{max} and P_{PV}^{min} are the upper and lower limits of the output of the distributed photovoltaic (PV); $\tan\phi_{PV}$ is the maximum output power factor of distributed PV; S_{PV}^{max} means the maximum capacity of distributed PV; P_{load} and Q_{load} are the active and reactive load at each node; $q_{C1/2/3/4}$ represents the per-unit reactive power compensation corresponding to different switching levels of the capacitor banks; and S_0 is the per-unit maximum capacity of the transformer.

TABLE RII
Values of Relevant Parameters for 38-node System (Unit: p.u.)

Name	Values	Name	Values	Name	Values
l_{min}	0.005	v_{MV}^{min}	0.95	S_{PV}^{max}	0.8
M	120	v_{MV}^{max}	1.05	$q_{RC1/2/3/4/5}$	-1/-0.5/0/0.5/1
ε	0.001	P_{PV}^{max}	0.8	S_c	1
S_{Line}	1.5	P_{PV}^{min}	0		
v_{OLTC}	0.98/1/1.02	$\tan\phi_{PV}$	1		

$q_{RC1/2/3/4/5}$ is the reactive power output of the reactor bank/ capacitor bank at the node. In addition, to make the results more pronounced, the load of the 38-node system is increased to five times its original level.

TABLE RIII

VALUES OF RELEVANT PARAMETERS FOR 162-NODE SYSTEM (UNIT: P.U.)

Name	Values	Name	Values	Name	Values
l_{min}	0.005	v_{MV}^{min}	0.95	S_{PV}^{max}	1
M	120	v_{MV}^{max}	1.05	S_{Line}	1
ε	0.001	P_{PV}^{max}	1	S_c	1
$q_{RC1/2/3/4/5}$	-1/-0.5/0/0.5/1	P_{PV}^{min}	0		
v_{OLTC}	0.98/0.99/1/1.01/1.02	$\tan\phi_{PV}$	1		

IV. THE PFR OF THE UNBALANCED MODIFIED SWISS 162-NODE TEST SYSTEM

The PFR of the unbalanced modified Swiss 162-node test system is also evaluated. Here, the estimated PFR is the PFR under the assumption of identical three-phase quantities, as shown in Fig. R2.

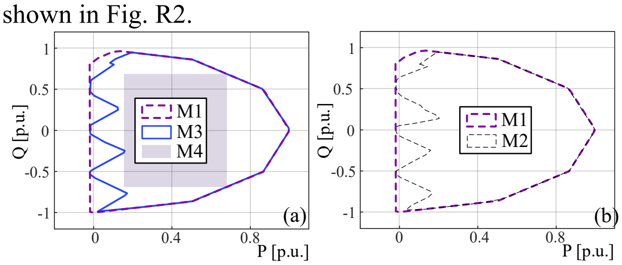

Fig. 2. Different PFRs by (a) M1, M3 and M4 for 162-node system; (b) M1 and M2 for 162-node system.

V. SUPPLEMENTARY DESCRIPTION OF THE RELEVANT PARAMETERS IN TABLE I

The intersection-over-union (IoU) is defined as the ratio of the overlap area between two regions to their total covered area. The infeasible inclusion ratio (IIR) and the feasible omission ratio (FOR) represent the proportions of infeasible operating points incorrectly included and feasible operating points incorrectly excluded, respectively.

Specifically, let A and B denote the estimated region 1 and the true region 2, respectively, and let $|\cdot|$ denote the area of a region. Then, the IoU, IIR and FOR can be calculated by (1)-(3).

$$IoU = \frac{|A \cap B|}{|A \cup B|}, \tag{1}$$

$$IIR = \frac{|A - A \cup B|}{|B|}, \tag{2}$$

$$FOR = \frac{|B - A \cup B|}{|B|}. \tag{3}$$

$IoU = 1$ indicates that the two regions completely overlap, whereas $IoU = 0$ indicates that they do not overlap at all. Therefore, the IoU serves as a measure of similarity between two regions: the closer the IoU is to 1, the higher the similarity; otherwise, the lower the similarity.

For the IIR and FOR, the closer the value is to 0, the fewer infeasible points are included and the fewer feasible points are missed.

V. SUPPLEMENTARY DESCRIPTION OF THE PROPOSED PFR IDENTIFICATION ALGORITHM

A. Proof of Convergence of the PFR Identification Algorithm

In Step 1, the algorithm determines two groups of binary variable combinations by solving (5), and in Step 3, it explores new combinations of the other variable set by (7) and (8). It should be noted that the constraint $l_m \geq l_{m,\mathrm{C}} + \varepsilon$ in (7) ensures that, unless a solution satisfying $l_{m,\mathrm{D}} = \min_{\boldsymbol{X}\in\Omega^1_{\mathrm{PFR}}(\boldsymbol{Z}_\mathrm{B}),\boldsymbol{Y},l_m} l_m$ in Step 2 is found, the integer variables obtained from (7) are different from all previously explored combinations of the other variable set. Once the binary variable combination $\boldsymbol{Z}_\mathrm{B}$ is explored again, the termination condition is met and the algorithm terminates. Since the number of integer variable combinations is finite, the algorithm must converge in a finite number of iterations. Let N_{01} denote the total number of binary variable combinations. Then, the worst-case number of iterations is N_{01}. In summary, the algorithm is guaranteed to converge within a finite number of iterations.

B. Sensitivity of the proposed algorithm to M, l_{min} and l_{tol}

In this section, the sensitivity of the proposed PFR identification algorithm to the number of angular partitions M, the gap for terminating the line-search iteration l_{min}, and the small positive size l_{tol} is evaluated on the 13-node test system. In the following tables, the IoU always denotes the intersection-over-union between the PFR obtained by the proposed method and that obtained by the enumeration method. The answers are as follows.

We first assess the sensitivity of the proposed algorithm to M. The test scenario is identical to the 13-node test system presented in the manuscript, except that only the value of M is changed.

TABLE RIV

THE SENSITIVITY OF THE PFR IDENTIFICATION ALGORITHM TO M

M	15	30	60	120	240	360
IoU	0.875	0.955	0.993	0.995	0.995	0.995

It can be observed that as M increases, the IoU first increases and then tends to remain unchanged. This indicates that the estimation accuracy of the PFR generally improves with the increase of M. However, beyond a certain threshold, for example when $M = 120$, the improvement in PFR accuracy becomes marginal and the IoU remains almost unchanged. Therefore, in practical applications, M should be properly selected to achieve a trade-off between evaluation accuracy and computational cost.

Then, the sensitivity of the proposed algorithm to l_{min}, and l_{tol} is evaluated. Except for the parameter under evaluation, all other settings are the same as those in the manuscript.

TABLE RV

TDI FLEXIBILITY UNDER DIFFERENT UNCERTAINTIES

l_{min}	l_{tol}	IoU
0.005	0.0001	0.995
	0.001	0.995
	0.01	0.995
	0.1	0.995
0.001	0.001	0.995
0.005		0.995
0.05		0.979
0.5		0.956

It can be observed that, in the test case, increasing l_{tol} has almost no impact on the IoU. This is because l_{tol} mainly affects the identification of interior infeasible points, whereas the tested case contains no interior infeasible points and only exhibits nonconvexity near the boundary. Therefore, the resulting impact is limited. In practical tests, it is still recommended that l_{tol} be set smaller than l_{min}. In addition, as l_{min} increases, the IoU continuously decreases. This is because l_{min} affects the identification of nonconvex boundaries and interior infeasible regions of the PFR. An excessively large l_{min} may cause feasible points to be incorrectly classified as infeasible during the PFR identification process, thereby reducing the IoU. Therefore, in the test case, a relatively small value of l_{min}, such as 0.005, is recommended.

α	σ_{PV}^2	σ_{Load}^2	M	
			12-node system	141-bus system
Deterministic			262.3	201.2
0.01	0.01	0.01	170.5	192.7
0.05	0.01	0.01	176.1	195.3
0.1	0.01	0.01	246.7	196.6
	0.05	0.01	167.6	183.6
	0.01	0.05	183.2	122.4

α	σ_{PV}^2	σ_{Load}^2	M	
			12-node system	141-bus system
Deterministic			262.3	201.2
0.01	0.01	0.01	170.5	192.7
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	0.05	0.01	167.6	183.6
	0.01	0.05	183.2	122.4